

Pricing Frictions, Tax Incidence and Welfare: Evidence and Theory

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Abstract

This paper explores how optimization frictions in supplier price-setting affect tax incidence. We use a natural experiment from the staggered adoption of Voluntary Collection Agreements (VCAs) by a major rental platform, which created an effective tax increase for hosts. On average, hosts reduced pre-tax prices by 5.2%, absorbing 43% of the new tax. However, this average effect masks significant heterogeneity. Pass-through is lowest for sophisticated hosts—defined as those whose price movements before the policy closely follow local hotels. We develop a model with heterogeneous price-setting frictions that shows how unsophisticated hosts “anchor” the market price to the previous equilibrium, thus shifting the tax burden to consumers. Counter-intuitively, this mechanism can lead to higher post-tax profits for all firms compared to a market without frictions.

Keywords: tax incidence, pricing frictions, tax compliance, salience

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1 Introduction

In classic models, tax incidence is fully determined by relative demand and supply elasticities (Myles, 1989; Weyl and Fabinger, 2013). Newer models, influenced by behavioral economics, reflect that individual consumers are influenced by a multitude of optimization frictions that affect relative tax burden in practice (Chetty, 2009; Finkelstein, 2009). However, in certain markets, *suppliers*—individuals with minimal price-setting experience—are affected by these same optimization frictions. How does this shape tax incidence and welfare?

We explore this question in the context of short-term rental markets using listings data from a major platform firm, Airbnb. Several American cities signed Voluntary Collection Agreements (VCAs) at different times between 2014 and 2016. Per these agreements, Airbnb collected hotel tax at the time of booking on behalf of hosts. Because few hosts remitted taxes prior to VCA implementation and nearly all complied afterward, the agreements resulted in a plausibly exogenous effective tax increase.

We have two main empirical findings. First, using daily property-level price data as well as commercial hotel data, we document substantial pricing frictions among suppliers in the Airbnb market: nearly half of all individual hosts adjust prices infrequently and show limited responsiveness to local demand conditions or competitor hotel rates. In addition, there is evidence that hosts employ heuristic pricing; greater than a third of all hosts list integer prices divisible by five. These rigidities stand in contrast to the more dynamic and flexible pricing strategies employed by hotels.

Second, we exploit plausibly exogenous variation in the specific timing of VCA adoption to measure the pass-through of the effective tax. We use a staggered difference-in-differences paired with double machine learning (DML)¹ to estimate the causal effect of VCA adoption on pre-tax pricing and consumer demand. With an average effective tax rate of 12%, hosts reduced pre-tax prices by 5.2% on average, offsetting roughly 43% of the tax. This magnitude roughly comports with Bibler et al. (2021), who use a similar dataset to infer levels of tax evasion among hosts prior to VCA adoption.

However, the more significant discovery lies in the heterogeneity of the effect by our proxy for

¹This method for causal inference corrects for bias in high dimensional datasets by using machine learning algorithms to flexibly model the relationship between confounding and treatment/outcome variables, then applies a technique based on the Frisch-Waugh-Lovell theorem to remove bias.

pricing friction. For each property listing in the period preceding the VCA, we generate a correlation between that property’s pricing and local hotel pricing. We find that adjustments varied sharply: “low friction” hosts whose price movements closely followed those of hotels absorbed a sizable part of the tax, resulting in lower incidence on consumers; “high friction” hosts, in contrast, passed through virtually the full tax onto consumers. These heterogeneous pricing responses translated into correspondingly heterogeneous effects on bookings, with reservation ratios falling by nearly twice as much for high-friction hosts.

Taken together, our results suggest that pass-through in markets with significant populations of suppliers who are inattentive, inappropriately anchored, or face other optimization frictions is likely to differ meaningfully from markets where all or most suppliers are traditional firms. This characterization is likely to apply to most amateur suppliers on digital platforms, but could also potentially apply beyond platforms to any peer-to-peer market. Although there are various studies that establish that consumers face optimization frictions that affect their responsiveness to changes in tax rate or tax administration (e.g., [Chetty et al., 2009](#); [Homonoff, 2018](#); [Lockwood and Taubinsky, 2017](#)), there is comparatively little evidence on whether similar optimization frictions also affect suppliers. Our findings are an emerging strand within the incidence literature that asserts that the nature of the firm is an underexplored determinant of the burden passed on to consumers (e.g., [Kopczuk et al., 2016](#); [Harju et al., 2015](#)).

To more formally understand the welfare consequences of supplier salience and other optimization frictions, we build a generalized model of firm pricing behavior. Building on [Anderson et al. \(2001\)](#) work on tax incidence with strategic complementarities in differentiated-product oligopoly, we adapt the [Calvo \(1983\)](#) pricing framework to explicitly model heterogeneous price-setting frictions across firms in an ad-valorem tax setting. Our model generates three novel predictions. First, rigid price-setters create an “anchoring effect” that helps hosts maintain a high market price and shifts tax incidence toward consumers. This mechanism is distinct from the elasticity-based determinants in [Weyl and Fabinger \(2013\)](#). Second, we establish a counterintuitive welfare ranking in which firms in frictional markets earn higher profits post-tax than in frictionless benchmarks, despite greater market inefficiency, complementing insights on overshifting ([Conlon and Rao, 2020](#)) while providing a new mechanism based on heterogeneous price rigidity. Third, we predict that the consumer tax burden is maximized immediately upon implementation and gradually shifts to

producers over time, offering testable implications for understanding incomplete and time-varying tax pass-through.

2 Policy Background and Data

In this section, we outline the policy change, specifically the VCA. We also describe our data.

2.1 Policy

The emergence of two-sided digital markets has reshaped the way goods and services are traded, challenging traditional approaches to regulation and taxation. Unlike conventional firms, which own inventory and control pricing directly, two-sided platforms serve as intermediaries, connecting decentralized suppliers with consumers through centralized algorithms, interfaces, and policies. This structure complicates enforcement: suppliers are often individuals lacking formal business infrastructure, and platforms, while not transacting directly, retain substantial influence over pricing, visibility, and user behavior.

Airbnb exemplifies this model in the short-term rental (STR) market. It enables individual property owners to list accommodations for short-term stays, often operating outside the regulatory frameworks governing traditional hotels. Hosts independently set prices and availability, but their decisions are shaped by the platform’s interface and tools—such as pricing suggestions and tax settings—that mediate how listings are presented to consumers. This combination of decentralized supply and platform-mediated pricing introduces new frictions for local governments seeking to enforce occupancy tax laws.

In response to widespread noncompliance with STR tax obligations, Airbnb began signing Voluntary Collection Agreements (VCAs) with local jurisdictions starting in 2014². Under these agreements, Airbnb collects local occupancy taxes at the point of booking and remits them directly to city governments, effectively shifting the compliance burden from individual hosts to the platform. In return, cities often granted concessions, such as waiving back taxes, limiting audit access to host-level data, or formalizing the legality of STR activity.

²Initially, Airbnb claimed that its hosts were not subject to these taxes, arguing that the transactions were non-commercial

In response, Airbnb began signing Voluntary Collection Agreements (VCAs) with local jurisdictions. These agreements authorized Airbnb to collect occupancy taxes at the time of booking and remit them directly to tax authorities, effectively shifting the burden from individual hosts to the platform. In return, cities often granted concessions, such as waivers of back taxes, limits on audit access to host-level data, or legalization of STRs under relaxed permitting standards. Between 2014 and 2016, Airbnb signed VCAs with over twenty cities, including several in our sample: Philadelphia, San Diego, Phoenix, Los Angeles, and Boulder.

While VCAs aim to improve tax collection, they also raise concerns about platform control, host pricing behavior, and market distortion. Many Airbnb hosts are individuals without formal business training or pricing expertise, and the centralization of compliance through the platform may further constrain their flexibility. These frictions—stemming from non-professional supply and platform-mediated decision-making—affect both the incidence of taxation and the structure of competition. They highlight growing regulatory concerns about the concentration of pricing power in digital platforms and reinforce calls for stronger oversight by agencies such as the FTC. In recent policy statements and enforcement actions, the FTC has increased scrutiny of hidden fees, pricing opacity, and algorithmic steering, including new rules aimed at restricting deceptive pricing practices in sectors like hospitality and ticketing.³

2.2 Data & Descriptive Statistics

Our analysis draws on two primary data sources, covering the period from December 2014 to August 2016. The main dataset is Airbnb listing-level data obtained from AirDNA, a third-party provider that collects and curates publicly available information from the Airbnb platform. These data offer high-frequency, property-level detail on nightly prices, availability, reservation activity, and listing characteristics across selected U.S. cities and their surrounding metropolitan areas.

To benchmark trends in the traditional accommodation sector, we supplement the Airbnb data with daily market-level hotel data from STR (Smith Travel Research). The STR dataset includes key hotel performance metrics—such as occupancy rates, average daily rates (ADR), and revenue per available room (RevPAR)—aggregated at the city or MSA level. These data enable direct comparisons

³See Federal Trade Commission (2023), “Policy Statement on Deceptive or Unfair Fees” and FTC’s 2024 proposed rule banning so-called “junk fees” in hotel, event, and platform pricing.

between digital platform rental activity and hotel market outcomes over the same period.

We focus on a balanced sample of ten U.S. cities: five that adopted Voluntary Collection Agreements (VCAs) during the sample period—Phoenix, San Diego, Boulder, Philadelphia, and Los Angeles—and five that did not—Houston, Dallas, Austin, New Orleans, and Savannah. These cities were selected based on data availability, consistent short-term rental activity, and comparable tourism intensity. The restriction to a smaller, well-observed set of cities facilitates a clean event-study design and minimizes confounding variation from staggered treatment timing or inconsistent data coverage.

To ensure that our sample captures active short-term rental properties, we impose several restrictions on the raw Airbnb data. First, we exclude extreme outliers in property size by dropping listings with more than five bedrooms or a maximum guest capacity exceeding twelve. These rare listings likely serve specialized or commercial functions and are not representative of typical Airbnb offerings. Second, we restrict the sample to active hosts by requiring that each listing have at least one reservation in the pre-policy period. This step removes approximately 30% of the original sample and ensures we focus on properties with genuine market activity. Finally, we limit the analysis to a short-run window around VCA implementation to isolate immediate behavioral responses. Specifically, we keep observations within a symmetric window spanning five weeks before and twelve weeks after the policy’s effective date.

Descriptive Statistics. [Table 1](#) summarizes the key characteristics of listings in treated and control cities. Columns 1 and 2 report the number of unique listings observed in each city’s MSA and within city limits, respectively. The remaining columns describe average listing attributes, including nightly price, number of bedrooms, and room type composition. Overall, listing characteristics are broadly comparable across treatment and control markets, providing support for the empirical strategy based on parallel trends.

3 Empirical Evidence of Host Pricing Frictions

In this section, we present descriptive empirical evidence which suggests Airbnb hosts face significant frictions in setting pre-tax prices. This evidence underlies our heterogeneity analysis in the

next section.

Airbnb hosts differ significantly from hotels and each other in several dimensions relevant to price setting: information access, experience, technologies, cognitive constraints, and adjustment costs. Large hotel chains frequently employ centralized pricing, optimized in corporate headquarters with advanced dynamic pricing algorithms that automatically adjust prices in response to minute changes in local demand. In contrast, most Airbnb hosts have minimal access to data relevant to price setting for their market, have little previous experience in pricing within the hospitality industry, and must manually change their default price for each date using the Airbnb Calendar.⁴

An emerging literature within industrial organization literature has sought to categorize pricing frictions in peer market pricing and identify their underlying mechanisms (Huang 2022, Pan 2019, For our purposes, however, we aim to establish the existence and extent of pricing frictions; we remain agnostic as to the precise mechanisms that give rise to the market-level pricing patterns we document.

Fact #1. Airbnb hosts rely on pricing heuristics. We first consider a "static" price setting friction, i.e. one that affects the selection of nominal price values. Figure 1 displays a histogram of listing prices under \$250. There is obvious bunching on integer values divisible by ten, as well as more moderate bunching at integer values divisible by five. It is unlikely that optimal prices reliably have this characteristic, suggesting that hosts are using heuristic pricing strategies akin to a rounding rule in increments of five. It is also notable that, in contrast to hotels, hosts seem unlikely to exploit the well documented left digit bias among consumers, which would present as bunching around price values ending in .99.

Fact #2. Most Airbnb hosts do not engage in dynamic pricing, or use simplified strategies. As tax incidence is determined by adjustments to the pre-tax price, frictions in price changes are highly relevant to our question. Figure 2 plots listing prices over time for sixteen randomly selected properties. Several of the properties, namely mini-plots 3,5,8,11 and 12, have perfectly horizontal price trajectories, indicating that the host never adjusted the initial price for the duration of the study

⁴During our study period, Airbnb provided very little guidance to hosts in initially setting prices, and no guidance on dynamic price adjustments because of concerns that providing such assistance might alter the legal characterization of hosts as independent contractors, thus exposing Airbnb to liability.

period. Miniplot 15 shows a single price change during the study period. Properties plotted in miniplots 1, 6, and 10 display a simplified dynamic pricing strategy in which price oscillates between two values, one value for weekdays and the other for weekend days. In contrast, miniplots 2, 4 and 7 show dynamic pricing more closely akin to what we might expect from a hotel operating with information about local demand shifts, with prices changing daily and exhibiting seasonal intercepts.

In fact, Airbnb hosts vary dramatically in the degree to which they face dynamic pricing frictions. Figure 3 introduces our preferred measure of pricing frictions; for each property, we calculate the correlation between average local daily hotel price series and the property’s price series. This measure has several desirable attributes. Unlike other proxies for frictions, such as the number of properties a host has, location, or duration of time on the platform, this measure is both dynamic and continuous, allowing for finer distinctions to be drawn between hosts. In addition, unlike other measures, it directly captures price setting acumen, if we make the reasonable assumption that hotel price movements represent profit maximizing adjustments.

Fact #3. Airbnb properties’ pre-policy correlation with local hotel prices is positively correlated with price adjustment after the policy. Our final descriptive fact concerns the relationship between pricing frictions and response to VCA implementation, a relationship we will return to in the next section. Figure 4 is a scatter plot showing a positive relationship between the correlation with hotel prices and the difference between pre policy and post policy average prices. While this shows the unrefined relationship between the two variables, we interpret it as providing initial evidence that hosts who face lower frictions in setting prices to maximize profit also face lower frictions in adjusting pre tax prices to share the hotel tax burden with consumers.

4 Empirical Analysis

4.1 The Effect of VCA on Host Price Adjustment

To identify the effect of VCA reform on host asking price adjustment, we employ the Difference-in-Differences (DiD) framework. The regression equation takes the following form:

$$\log(P_{ijt}) = \beta \cdot VCA_{jt} + \lambda_i + X_{jt} + \varepsilon_{ijt} \quad (1)$$

, where P_{ijt} is the consumer paid the price for property i in city j at year-week t , defined as the final price paid by the consumer, which includes the listed price plus the cleaning fee and the service fee. Note that the observed booking price should be tax-inclusive before a VCA is implemented but tax-exclusive afterwards. The main independent variable VCA_{jt} is a dummy variable indicating the VCA agreement is implemented in cities j at time t . This variable is equal to 1 if the VCA agreement is implemented at city j at time t , and equal to 0 if the VCA agreement is not implemented. We include property fixed effects λ_i to control for observed and unobserved time-invariant property-specific characteristics. We also include the metro-week covariance X_{jt} to capture the metro-specific seasonality demand shock. Finally, ε_{ijt} is the error term. All estimations are clustered at the MSA level. To address the potential concern due to the multiple-time treatment (Goodman-Bacon, 2021), we adopt a staggered version of difference-in-differences and use the never-treated cities as a control.

Table 2 shows the results for the log listed price from the host. Column (1) offers a basic DID estimate from a two-way fixed effect, and column (2) adds year-week cities demand as an additional control to capture city-specific seasonal demand change. Columns (3) and (4) use the staggered DID following Sun and Abraham (2021) and Callaway and Sant’Anna (2021). The point estimates remain consistently similar across all specifications and different versions of DiD, with Column (5) being our preferred estimate. Following Roth (2024), we estimate equation 1 in log-difference for both pre-VCA and post-VCA coefficients so we can directly interpret the results as traditional difference-in-differences throughout the paper. It suggests that the host lowers the listed price by 5.2% when the VCA is in place. Considering that the average tax rate is 12%, this indicates the host’s listed price decreased by 43% of the enforced tax. Panel B and Panel C further consider the heterogeneity and show the results of the property with bottom 50% price correlation (high price friction group) and with top 50% price correlation (low price friction group). The results align with the theoretical predictions: we find that the estimated effect on the host listed price is smaller for listing with high price friction (with price correlation below the median). The estimated effect of VCA on log listed price is -0.14 and is significant at 1%. In contrast, the estimated effect VCA for a property with lower price friction is -0.89 and is significant at 1%. It indicates that in reality, when facing the same average tax rate change, the price elasticity is a function of price friction.

Acknowledging that using the median correlation as a threshold to classify high and low price friction is somewhat arbitrary, we investigate the heterogeneous effects of VCA reform on listing

price adjustment across listings with different price correlation percentiles. More specifically, we estimate equation 3 for listings with different price frictions, categorizing by the price correlation during a month before the VCA reform. Figure 5 depicts the estimated VCA effect on both the log listed price and the consumer paid price for the property with price correlation within percentile X. The y-axis displays the estimated effect for a property with price correlation within the percentile X and its respective 95% confidence intervals. The x-axis indicates the within percentile X. We find that the estimated effect of VCA on both the log listed price and the consumer paid price intensifies as X increases. This pattern is consistent with the theoretical prediction on the role of price friction in determining host price adjustment since a higher value of X represents a higher price correlation with local professional hotel, and suggests a lower price frictions, as they can adjust price more rapidly and with greater acumen than properties with larger price friction in response to the tax change caused by the VCA reform.

In order to examine the common trend assumption and outline the full dynamic trajectory of the effects of VCA reform, we implement a dynamic difference-in-differences specification:

$$\log(Y_{ijt}) = \sum_{k=-5}^{k=12} \beta_k \cdot VCA_j \times 1[t = c + k] + X_{jt}\psi + \epsilon_{ijt}. \quad (2)$$

, where $1(y_{ijt} = c + k)$ is a dummy variable indicating the number of relative weeks to the VCA reform, where k does not include -1 since we consider a week before the reform as the reference year. β_k is the coefficient of interest, which captures the dynamic effects of the VCA reform on outcome relative to the reference year.

Figure 6 highlights the estimated β_k in Equation (2) using our preferred estimator from Callaway and Sant’Anna (2021). The horizontal axis represents the number of weeks relative to the VCA reform, and the figure includes 95% confidence intervals for these estimates. The black triangle dot represents the estimation for all properties, the blue dot stands for properties with the bottom 50% correlation, and the orange dot represents properties with the top 50% correlation.

Two key insights emerge from the dynamic DID analysis. First, there is no evidence of a pre-existing trend, as the coefficients are close to zero and statistically insignificant for both outcomes. This confirms that the average log listed price and log consumer paid price between treated and untreated groups were parallel before the VCA reform, validating the parallel trends assumption in

our DID design. Second, the impact of the VCA reform on host price adjustment is initially small and non-significant in the first week after the reform. However, the effect increases and becomes statistically significant after two weeks. Listings in the top 50% price correlation group reduce prices immediately after the first week, whereas listings in the bottom 50% correlation group adjust their prices more gradually.

We further show that the results for log consumer paid are driven by the change of host listed price but not the change in cleaning fee in [Table A2](#). Additionally, the results considering only the traded consumer paid price show similar results in [Table A3](#).

4.2 The Effect of VCA on Reservation Status

Building on the finding that the effect of VCA reform on consumer-paid price listed on Airbnb increases significantly based on host’s price friction, a pertinent question emerges: Will the corresponding reservation (quantity trade) alter in response to the different levels of price increase? To answer this question, we further examine the impact of VCA reform on reservation status using the following dynamic DID specification:

$$R_{ijt} = \beta \cdot VCA_{jt} + \lambda_i + X_{jt} + \varepsilon_{ijt} \quad (3)$$

The main difference between this specification and equation (1) is the outcome variable. Here, R_{ijt} is a log reservation ratio per week, which is calculated as the number of days booked divided by the number of days listed available on the Airbnb platform. The coefficients β_k on the VCA_{jt} capture the effect of VCA reform.

[Table 3](#) presents the results for the weekly quantity traded. Following the structure of [Table 2](#), Column (5) represents our preferred specification using the estimator from [Callaway and Sant’Anna \(2021\)](#) with a log-difference specification following [Roth \(2024\)](#). Panel A shows the effect on all listings, while Panels B and C detail the effects on listings with bottom and top 50% price correlation, respectively. Consistent with theoretical predictions, the estimated effect of VCA on log reservations significantly decreases for all properties, but the magnitude is larger for the listing with larger price friction due to a larger increase in the consumer-paid price. Specifically, the estimated effect of VCA on the reservation ratio is -0.105 and statistically significant, indicating a 10.5% decline in the

probability of a listed property being booked. Furthermore, this decrease in reservations is primarily driven by properties with above the median price correlation, where consumer paid prices increase the most. For properties below the median price correlation, the estimated effect of VCA on the reservation ratio is -0.124 and statistically significant, nearly twice as large as the effect for properties with top 50% price correlation (point estimate -0.091 and not statistically significant).⁵

To fully investigate the heterogeneous effects of the VCA reform in listings with different levels of price friction, we estimate equation 3 for listings categorized by price correlation during the month before the VCA reform. Figure 7 shows the estimated β for each listing with a price correlation within percentile X, along with their respective 95% confidence intervals. We find that the estimated effect of VCA on reservation intensifies as the listing's price correlation decreases (X decreases). This pattern is consistent with basic principles on the reservation in response to the price adjustment: the greater the price increase, the less quantity sold. The estimated effect of VCA on log reservation is larger for the listing in a lower price correlation percentile, as the larger the price friction, the higher the consumer paid price, and the lower the quantity traded. It suggests a clear welfare implication by considering listing with heterogeneous price friction in reality: listing with larger price friction cannot adjust the price as quickly/fast as the property without the price friction (in the traditional model without price friction), and their delayed price adjustment causes the quantity sold to decline more than the case without price friction.

Similarly, to examine the common trend assumption and outline the full dynamic trajectory of the effect of VCA on reservation, we estimate the dynamic difference-in-differences specification:

$$R_{ijt} = \sum_{k=-5}^{k=12} \beta_k \cdot VCA_j \times 1[t = c + k] + X_{jt}\psi + \epsilon_{ijt}. \quad (4)$$

Figure 8 presents the estimated β_k coefficients from equation (4) for all listings in black dots, listings with bottom 50% price correlation in blue triangles, and listings with top 50% price correlation in orange diamonds, respectively. The horizontal axis represents the number of weeks relative to the week of VCA reform, and the figures include 95% confidence intervals for these estimates. The black triangle dot represents the estimation for all properties, the blue dot stands for properties with the bottom 50% correlation, and the orange dot represents properties with the top 50% correlation.

⁵In the Appendix, we further validate that the number of listings is not changed by the VCA reform to support the idea that the change in quantity traded is driven by the price adjustment but not the number of listings available.

Two key insights emerge from the dynamic DID analysis. First, there is no evidence of a pre-existing time trends, as the coefficients are close to zero and statistically insignificant before a VCA is in effect. This confirms that the average log reservation between treated and untreated groups was parallel before the VCA reform, validating the parallel trends assumption in our DID design. Second, the impact of the VCA reform on reservation intensity decreases gradually for the first four weeks, and stays stable.

4.3 Robustness Checks

Online Appendix shows a series of robustness checks for our main findings. Appendix X shows the estimation using a different city demand proxy as control. Appendix B shows the estimation from the balanced sample.

5 Theoretical Model

In this section, we present a framework for modeling the effect of VCA on Airbnb market prices and tax incidence, where consumers are fully rational optimizers and hosts face pricing frictions.

We consider a market for differentiated properties populated by a continuum of hosts. Hosts are heterogeneous in their price-setting behavior and are categorized into three types. The introduction of a government policy is modeled as a new ad-valorem tax, τ , on host prices.

We show that the presence of rigid-price firms acts as an 'anchor,' shifting the long-run tax burden from producers to consumers. This mechanism leads to a counterintuitive welfare ranking where both flexible and rigid firms in a market with frictions can earn higher profits than firms in a frictionless benchmark.

5.1 Setting

The proportion of each host type is given by α_i , where:

- **Type 1: Rigid hosts** (proportion α_1): These hosts maintain a fixed price, p_0 , which was the equilibrium price before the introduction of the tax.

- **Type 2: Partially Flexible hosts** (proportion α_2): These hosts follow a Calvo-style (1983) pricing rule. In any given period, they can re-optimize their price with probability $\mu_2 \in (0, 1)$. With probability $1 - \mu_2$, they are constrained to charge the price they set at their last adjustment.
- **Type 3: Fully Flexible hosts** (proportion α_3): These hosts can and do re-optimize their price in every period.

The proportions sum to unity: $\sum_{i=1}^3 \alpha_i = 1$.

5.1.1 Cost

Regardless of their types, hosts have an identical cost function

$$C(q_j)$$

, with positive and increasing marginal costs (i.e., $C'(q_j) > 0$, $C''(q_j) \geq 0$), where q_j denotes the quantity of host j .

5.1.2 Demand

The quantity demanded for host j , q_j , is a function of its own consumer price, p_j^c , and the average consumer price, $\bar{p}^c$.

$$q_j = D(p_j^c, \bar{p}^c) \tag{5}$$

We assume demand is downward sloping in own-price ($\frac{\partial D}{\partial p_j^c} < 0$) and increasing in the average market price ($\frac{\partial D}{\partial \bar{p}^c} > 0$), as products are substitutes.

5.1.3 Profit

The government enacts an *ad valorem* tax, τ , so $p_j^c = p_j(1 + \tau)$, where p_j denotes the host price. The host's profit, Π , is its revenue minus its costs, which can be expressed as a general function of its own host price p_j , the average host price $\bar{p}$, and the tax rate τ .

$$\Pi_j = \Pi(p_j, \bar{p}, \tau) = p_j \cdot D(p_j(1 + \tau), \bar{p}(1 + \tau)) - C(D(p_j(1 + \tau), \bar{p}(1 + \tau))) \tag{6}$$

Our results rely on four standard assumptions about this profit function.

Assumption 1 (Concavity). The profit function is strictly concave in p_j , ensuring a unique optimal price.

$$\frac{\partial^2 \Pi}{\partial p_j^2} < 0$$

Assumption 2 (Strategic Complementarity (Supermodularity)). A host's marginal profitability from increasing its own price is increasing in the average market price. This is the model's core assumption.

$$\frac{\partial^2 \Pi}{\partial p_j \partial \bar{p}} > 0$$

Assumption 3 (Positive Environmental Effect:). A host's profit is increasing in the average market price, as higher rival prices increase its own demand.

$$\frac{\partial \Pi}{\partial \bar{p}} > 0$$

Assumption 4 (Tax Effect (Submodularity)). An increase in the tax rate τ decreases the marginal profitability of raising the *host* price p_j .

$$\frac{\partial^2 \Pi}{\partial p_j \partial \tau} < 0$$

5.1.4 Flexible Host's Problem

We assume the market is in equilibrium where all hosts charge p_0 ($\frac{\partial \Pi}{\partial p_j}(p_0, p_0, 0) = 0$) before the tax is enacted. After the tax is introduced, rigid hosts will persistently set their host price equal to the pre-tax price, p_0 . We assume flexible and semi-flexible hosts have backward-looking adoptive expectations⁶ and use the previous period's average price to form their expectations.

Then, a flexible host (Type 3, or Type 2 that gets to update) chooses its price p_t to solve:

$$\max_{p_t} \Pi(p_t, \bar{p}_{t-1}, \tau)$$

The first-order condition $\frac{\partial \Pi}{\partial p_t}(p_t, \bar{p}_{t-1}, \tau) = 0$ implicitly defines the optimal price p_t^* as a general best-

⁶We employ backward-looking adaptive expectations for both theoretical and empirical reasons. First, this framework reflects a plausible form of bounded rationality for the producers in this specific market (i.e., individual hosts), who are more likely to use simple heuristics based on recently observed market prices than to solve a complex, forward-looking dynamic optimization problem. Second, this assumption endogenously generates the gradual price adjustment and inertia that is commonly observed in high-frequency price data following a shock, which aligns well with our empirical investigation.

response function:

$$p_t^* = p^*(\bar{p}_{t-1}, \tau) \quad (7)$$

Based on our assumptions on the profit function, the optimal price has two key properties: (i) it increases in the average price, i.e., $\frac{\partial p^*}{\partial \bar{p}_{t-1}} > 0$ ⁷. which is a direct result of strategic complementarity, (ii) it decreases in the tax rate, $\frac{\partial p^*}{\partial \tau} < 0$.

5.2 Law of Motion

We can also characterize the law of motion for the average price. The average market price at time t , $\bar{p}_t$, aggregates the prices of all hosts. Using the standard Calvo definition (where non-updating Type 2 hosts keep their prior-period price, $p_{2,t-1}$, the price is:

$$\bar{p}_t = \alpha_1 p_0 + \alpha_3 p_t^* + \alpha_2 p_{2,t} \quad (8)$$

where $p_{2,t}$ is the average price of Type 2 hosts, which evolves as:

$$p_{2,t} = (1 - \mu_2)p_{2,t-1} + \mu_2 p_t^* \quad (9)$$

Solving for $p_{2,t-1}$ in the $t - 1$ version of the first equation and substituting it into the second. yields a general, non-linear, second-order difference equation:

$$\bar{p}_t = \alpha_1 \mu_2 p_0 + (1 - \mu_2)\bar{p}_{t-1} + (\alpha_3 + \alpha_2 \mu_2)p_t^* - (1 - \mu_2)\alpha_3 p_{t-1}^* \quad (10)$$

Substituting the best-response function $p_t^* = p^*(\bar{p}_{t-1}, \tau)$ gives the full law of motion:

$$\bar{p}_t = \alpha_1 \mu_2 p_0 + (1 - \mu_2)\bar{p}_{t-1} + (\alpha_3 + \alpha_2 \mu_2)p^*(\bar{p}_{t-1}, \tau) - (1 - \mu_2)\alpha_3 p^*(\bar{p}_{t-2}, \tau) \quad (11)$$

This equation shows that the current average price $\bar{p}_t$ depends on the last two periods ($\bar{p}_{t-1}$ and $\bar{p}_{t-2}$), giving the model its dynamic persistence, or "memory."

We can also characterize the long-run equilibrium. As $t \rightarrow \infty$, the market settles into a steady state where $\bar{p}_t = \bar{p}_{t-1} = \bar{p}_\infty$ and $p_t^* = p_\infty^*$. In this equilibrium, all partially flexible hosts (Type 2) have had a chance to update, so their prices converge to the same optimal price p_∞^* as the flexible

⁷We assume $\frac{\partial p^*}{\partial \bar{p}_{t-1}} < 1$ for dynamic stability.

hosts (Type 3). The long-run price is therefore a weighted average of only the rigid hosts and all non-rigid hosts $(1 - \alpha_1)$: The dynamic adjustment path (which is non-linear in this general model) converges to this long-run equilibrium.

This equilibrium is defined by the solution to the following two-equation system:

$$p_\infty^* = p^*(\bar{p}_\infty, \tau) \quad (12)$$

$$\bar{p}_\infty = \alpha_1 p_0 + (1 - \alpha_1) p_\infty^* \quad (13)$$

Special Cases.

- **Frictionless** ($\alpha_1 = 0$): The Aggregation equation (13) becomes $\bar{p}_\infty = p_\infty^*$. Plugging this into the Best-Response equation (12) gives $p_\infty^* = p^*(p_\infty^*, \tau)$, which is the definition of the frictionless equilibrium p^* . Thus, $\bar{p}_\infty(0) = p^*$.
- **Fully Rigid** ($\alpha_1 = 1$): The Aggregation equation (13) becomes $\bar{p}_\infty = p_0$. Thus, $\bar{p}_\infty(1) = p_0$.

5.3 Model Results

In this section, we will present the results on welfare and tax burden. We also compare the friction model (where $\alpha_1 > 0$) to the frictionless benchmark (where $\alpha_1 = 0$).

In the frictionless world, in the absence of a tax ($\tau = 0$), a host solves $\max_{p_j} \Pi(p_j, \bar{p}, 0)$. In a symmetric equilibrium, $p_j = \bar{p} = p_0$. This pre-tax equilibrium price p_0 is the unique solution to the first-order condition:

$$\frac{\partial \Pi}{\partial p_j}(p_0, p_0, 0) = 0 \quad (14)$$

This p_0 is the price that "rigid" firms are constrained to use post-tax in the friction world.

When an ad-valorem tax is introduced, all hosts re-optimize. A firm solves $\max_{p_j} \Pi(p_j, \bar{p}, \tau)$. In the new symmetric equilibrium, $p_j = \bar{p} = p^*$. This frictionless post-tax price p^* is the solution to:

$$\frac{\partial \Pi}{\partial p_j}(p^*, p^*, \tau) = 0 \quad (15)$$

From our Tax Effect assumption (Assumption 4), it follows that the new optimal producer price is lower than the pre-tax price: $p^* < p_0$.

5.3.1 Tax Burden

Proposition 1 (Tax Burden Shift). As the proportion of rigid firms (α_1) increases, the Producer Tax Burden (PTB) decreases and the Consumer Tax Burden (CTB) increases.

Proof. As defined in the tax literature $PTB = p_0 - \bar{p}_\infty$ and $CTB = \bar{p}_\infty(1 + \tau) - p_0$. The proof rests on showing that $\bar{p}_\infty$ is an increasing function of α_1 . The steady state is implicitly defined by Equation 13::

$$\bar{p}_\infty = \alpha_1 p_0 + (1 - \alpha_1) p^*(\bar{p}_\infty, \tau)$$

Let $F(\bar{p}_\infty, \alpha_1) = \bar{p}_\infty - \alpha_1 p_0 - (1 - \alpha_1) p^*(\bar{p}_\infty, \tau) = 0$. By the Implicit Function Theorem (IFT), we know

$$\frac{d\bar{p}_\infty}{d\alpha_1} = -\frac{\partial F / \partial \alpha_1}{\partial F / \partial \bar{p}_\infty} \quad (16)$$

The partial derivatives are:

- $\frac{\partial F}{\partial \alpha_1} = p_\infty^* - p_0$
- $\frac{\partial F}{\partial \bar{p}_\infty} = 1 - (1 - \alpha_1) \frac{\partial p^*}{\partial \bar{p}_\infty}$

Substituting these into Equation 16 gives:

$$\frac{d\bar{p}_\infty}{d\alpha_1} = \frac{p_0 - p_\infty^*}{1 - (1 - \alpha_1) \frac{\partial p^*}{\partial \bar{p}_\infty}} \quad (17)$$

By the supermodularity assumption (Assumption 2) $0 < \frac{\partial p^*}{\partial \bar{p}_\infty} < 1$. Thus, the denominator is $1 - (\text{a value} < 1)$, which is positive. Therefore, to show that Equation 17 is positive, we must show $p_0 > p_\infty^*$. By definition, $p_0 = p^*(p_0, 0)$ and $p_\infty^* = p^*(\bar{p}_\infty, \tau)$. Since $\bar{p}_\infty \leq p_0$ and $\tau > 0$, by Assumption 2 we have,

$p_\infty^* = p^*(\bar{p}_\infty, \tau) \leq p^*(p_0, \tau)$. Then by the tax effect (Assumption 4) $p^*(p_0, \tau) < p^*(p_0, 0) = p_0$. Thus, $p_0 - p_\infty^*$ is **positive**. Therefore, Equation 17 is positive and $\bar{p}_\infty$ is strictly increasing in α_1 .

Since $\frac{d\bar{p}_\infty}{d\alpha_1} > 0$, we can conclude that:

- $\frac{d(PTB)}{d\alpha_1} = -\frac{d\bar{p}_\infty}{d\alpha_1} < 0$. (Producer burden decreases)
- $\frac{d(CTB)}{d\alpha_1} = (1 + \tau) \frac{d\bar{p}_\infty}{d\alpha_1} > 0$. (Consumer burden increases)

□

Corollary 1 (Producer Tax Burden Comparison). For any proportion of rigid firms $\alpha_1 > 0$, the long-run Producer Tax Burden in the friction model ($PTB_{friction}$) is strictly lower than the Producer Tax Burden in the standard frictionless model ($PTB_{standard}$).

Proof. Let $PTB_{standard} = p_0 - p^*$, $PTB_{friction} = p_0 - \bar{p}_\infty$. As established in the proof of Proposition 1, the long-run average price $\bar{p}_\infty$ is a strictly increasing function of α_1 , with a lower bound of p^* at $\alpha_1 = 0$. Therefore, for any $\alpha_1 > 0$:

$$\bar{p}_\infty > p^*$$

Since $\bar{p}_\infty > p^*$, it follows that $-\bar{p}_\infty < -p^*$. Adding p_0 to both sides of the inequality gives:

$$p_0 - \bar{p}_\infty < p_0 - p^*$$

Therefore, $PTB_{friction} < PTB_{standard}$. The producers in the friction model bear a strictly smaller tax burden. $\square$

5.3.2 Profits

Proposition 2 (Welfare Ranking of Firms). In the post-tax, long-run equilibrium, flexible firms earn strictly higher profits and suffer a strictly smaller welfare loss than rigid firms.

Proof. Both firm types operate in the same market environment $(\bar{p}_\infty, \tau)$. A flexible firm chooses $p_\infty^* = \arg\max_{p_j} \Pi(p_j, \bar{p}_\infty, \tau)$. A rigid firm is constrained to p_0 . As shown in Prop. 1, $p_0 \neq p_\infty^*$, so p_0 is suboptimal. By definition of maximization and concavity (Assumption 1):

$$\Pi_{flex} = \Pi(p_\infty^*, \bar{p}_\infty, \tau) > \Pi(p_0, \bar{p}_\infty, \tau) = \Pi_{rigid}$$

$\square$

Proposition 3 (Benefit of Frictions for Flexible Firms). A flexible firm in the friction model (with $\alpha_1 > 0$) earns a strictly higher profit than a firm in the frictionless benchmark model.

Proof. Let $\Pi_{max}(\bar{p}, \tau)$ be the flexible firm's value function. By the Envelope Theorem, the derivative of this function with respect to the average market price is:

$$\frac{d\Pi_{max}}{d\bar{p}} = \frac{\partial \Pi}{\partial \bar{p}}$$

By Assumption 3 (Positive Environmental Effect), $\frac{\partial \Pi}{\partial \bar{p}} > 0$. Thus, Π_{max} is a strictly increasing function of the average market price $\bar{p}$.

From the proof of Proposition 1, $\bar{p}_\infty(\alpha_1)$ is strictly increasing in α_1 with the lower bound being $\bar{p}_\infty(0) = p^*$. Therefore, for any $\alpha_1 > 0$, the average price $\bar{p}_\infty$ is strictly greater than the frictionless price p^* .

$$\bar{p}_\infty > p^* \quad (\text{for } \alpha_1 > 0)$$

Since Π_{max} is strictly increasing in the average price, and the average price is strictly higher in the friction model, the profit must also be strictly higher.

$$\Pi_{flex} = \Pi_{max}(\bar{p}_\infty, \tau) > \Pi_{max}(p^*, \tau) = \Pi^*$$

□

For the remaining propositions, we will use the following lemma, which states that the average producer price in the market with frictions ($\bar{p}_\infty$) falls more slowly (or less) than the producer price in the standard frictionless market (p^*).

Lemma 1 (The Anchoring Effect). The average producer price in the market with frictions ($\bar{p}_\infty$) falls more slowly (or less) than the producer price in the standard frictionless market (p^*).

Proof. We need to show that at $\tau = 0$, the derivative of the average producer price with respect to the tax is strictly greater (less negative) than the derivative of the frictionless price, for any $\alpha_1 > 0$.

$$\left. \frac{d\bar{p}_\infty}{d\tau} \right|_{\tau=0} > \left. \frac{dp^*}{d\tau} \right|_{\tau=0}$$

For a small tax, the long-run equilibrium optimal price p_∞^* can be approximated by the frictionless price $p^*(\tau)$. The aggregation equation is:

$$\bar{p}_\infty(\tau) = \alpha_1 p_0 + (1 - \alpha_1) p_\infty^*(\tau) \approx \alpha_1 p_0 + (1 - \alpha_1) p^*(\tau)$$

Differentiate this approximation with respect to τ (noting p_0 is a constant):

$$\frac{d\bar{p}_\infty}{d\tau} \approx (1 - \alpha_1) \frac{dp^*}{d\tau}$$

Substitute this into the difference equation, $G'_\tau(0) = \left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau} \right) \Big|_{\tau=0} :$

$$G'_\tau(0) \approx \left((1 - \alpha_1) \frac{dp^*}{d\tau} - \frac{dp^*}{d\tau} \right) \Big|_{\tau=0} = -\alpha_1 \frac{dp^*}{d\tau} \Big|_{\tau=0}$$

Note that by Assumption 4 (Tax Effect), $\frac{dp^*}{d\tau}\big|_{\tau=0} < 0$ and since $\alpha_1 > 0$, $G'_\tau(0) > 0$

□

Proposition 4 (Welfare of Rigid Firms). For a sufficiently small tax ($\tau > 0$), a rigid firm in the friction model earns a higher profit than a firm in the frictionless benchmark ($\Pi_{rigid} > \Pi^*$) as long as there is any positive proportion of rigid firms in the market ($\alpha_1 > 0$).

This outcome is driven by two competing effects: (1) a negative "suboptimal pricing effect" from being constrained to p_0 , and (2) a positive "environmental effect" from other rigid firms raising the average price $\bar{p}_\infty$. For a small tax, the price deviation is small, making the loss a second-order effect. The environmental gain, however, is a first-order effect. The positive first-order effect dominates.

Proof. Let $\Delta\Pi(\tau) = \Pi_{rigid}(\tau) - \Pi^*(\tau)$. At $\tau = 0$, $\Delta\Pi(0) = 0$. By chain rule:

$$\frac{d\Pi_{rigid}}{d\tau} = \frac{\partial\Pi}{\partial\bar{p}_\infty} \frac{d\bar{p}_\infty}{d\tau} + \frac{\partial\Pi}{\partial\tau} \quad (18)$$

By Envelope Theorem

$$\frac{d\Pi^*}{d\tau} = \frac{\partial\Pi}{\partial\bar{p}}\bigg|_{p^*} \frac{dp^*}{d\tau} + \frac{\partial\Pi}{\partial\tau}\bigg|_{p^*} \quad (19)$$

The derivative of the profit difference at $\tau = 0$ is as follows

$$\frac{d(\Delta\Pi)}{d\tau}\bigg|_{\tau=0} = \left(\frac{d\Pi_{rigid}}{d\tau} - \frac{d\Pi^*}{d\tau} \right)\bigg|_{\tau=0} \quad (20)$$

and at $\tau = 0$, $\bar{p}_\infty = p^* = p_0$, so all partial derivatives $\frac{\partial\Pi}{\partial\bar{p}}$ and $\frac{\partial\Pi}{\partial\tau}$ are identical.

$$\frac{d(\Delta\Pi)}{d\tau}\bigg|_{\tau=0} = \left(\frac{\partial\Pi}{\partial\bar{p}} \frac{d\bar{p}_\infty}{d\tau} + \frac{\partial\Pi}{\partial\tau} \right) - \left(\frac{\partial\Pi}{\partial\bar{p}} \frac{dp^*}{d\tau} + \frac{\partial\Pi}{\partial\tau} \right) \quad (21)$$

$$\frac{d(\Delta\Pi)}{d\tau}\bigg|_{\tau=0} = \frac{\partial\Pi}{\partial\bar{p}}\bigg|_{\tau=0} \cdot \left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau} \right)\bigg|_{\tau=0} \quad (22)$$

Recall that $\frac{\partial\Pi}{\partial\bar{p}} > 0$ by Assumption 3 and $\left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau} \right)\bigg|_{\tau=0} > 0$ by Lemma 1. Since $\Delta\Pi(0) = 0$ and $\Delta\Pi'(0) > 0$, $\Delta\Pi(\tau)$ must be positive for a sufficiently small $\tau > 0$. □

Corollary 2 (The Complete Post-Tax Welfare Ranking). For a sufficiently small tax ($\tau > 0$), there is

a strict welfare (profit) ordering:

$$\Pi_{flex} > \Pi_{rigid} > \Pi^*$$

Proof. This corollary summarizes the previous three general-model proofs.

- $\Pi_{flex} > \Pi_{rigid}$ (from Proposition 2).
- $\Pi_{rigid} > \Pi^*$ (from Proposition 4, for small τ).
- $\Pi_{flex} > \Pi^*$ (from Proposition 3).

□

5.3.3 Excess Burden

For the following proposition, we need to be able to compare equilibrium quantities. To do so, we add a standard assumption. Aggregate market quantity $Q_{agg}(p^c)$ is a strictly decreasing function of the symmetric consumer price p^c .

Proposition 5 (The Inefficiency of Price Frictions). For any $\tau > 0$ and $\alpha_1 > 0$, the Deadweight Loss in the friction model ($DWL_{friction}$) is strictly greater than the Deadweight Loss in the standard frictionless model ($DWL_{standard}$).

Proof. We prove this by showing that the friction model has a higher tax wedge and a lower quantity.

The wedges are $\tau \bar{p}_\infty$ and τp^* . From Proposition 3, we proved that for any $\alpha_1 > 0$, $\bar{p}_\infty > p^*$. Since $\tau > 0$, the tax wedge is strictly larger: $(\tau \bar{p}_\infty) > (\tau p^*)$.

The long-run average consumer prices in both settings are as follows: $p_c^{flex} = \bar{p}_\infty(1 + \tau)$ and $p_c^* = p^*(1 + \tau)$. Since $\bar{p}_\infty > p^*$, it is clear $p_c^{flex} > p_c^*$. By the assumption we made earlier, Q_{agg} is a strictly decreasing function. Therefore:

$$q_{avg} = Q_{agg}(p_c^{flex}) < Q_{agg}(p_c^*) = q^*$$

The average quantity in the friction model is strictly smaller, implying a strictly larger reduction in quantity from the pre-tax level.

The friction model has both a strictly larger tax wedge and a strictly larger reduction in quantity, hence, $DWL_{friction} > DWL_{standard}$

□

5.3.4 Comparative Statics

Proposition 6 (Flexible Firm vs. Standard Case). The profit advantage that a flexible firm holds over a firm in the standard model, $\Delta\Pi_{flex} = \Pi_{flex} - \Pi^*$, is a strictly increasing function of the tax rate τ .

Proof. As proven in Proposition 3, $\Pi_{max}(\bar{p}, \tau)$ is strictly increasing in $\bar{p}$. Define $G_p(\tau) = \bar{p}_\infty(\tau) - p^*(\tau)$. This gap is positive for $\alpha_1 > 0$, and as proven in Lemma 1, this difference is positive. The profit advantage is $\Delta\Pi_{flex} = \Pi_{max}(\bar{p}_\infty, \tau) - \Pi_{max}(p^*, \tau)$. Since Π_{max} is an increasing function and its argument's advantage, $G_p(\tau)$, is also increasing in τ , the profit advantage $\Delta\Pi_{flex}$ must be a strictly increasing function of τ . □

Proposition 7 (Non-Monotonicity of Rigid Firm Profit). The effect of the tax rate on the profit difference $\Delta\Pi_{rigid} = \Pi_{rigid} - \Pi^*$ initially increases for small taxes, but must eventually decrease for larger taxes.

Proof. We need to show that

1. The difference is initially increasing: $\frac{d(\Delta\Pi_{rigid})}{d\tau}\big|_{\tau=0} > 0$.
2. The difference must eventually decrease, as $\lim_{\tau \rightarrow \infty} \Delta\Pi_{rigid}(\tau) = 0$.

Since the function starts at 0, increases, and must return to 0, there must exist at least one threshold $\tilde{\tau} > 0$ such that for all $\tau > \tilde{\tau} > 0$, the function is decreasing.

1. As proven in Proposition 4, the positive "environmental" effect (a first-order gain from $\bar{p}_\infty > p^*$) dominates the "suboptimal pricing" effect (a second-order loss from p_0 being near the optimum) hence the derivative of the profit difference at $\tau = 0$ is strictly positive:

$$\frac{d(\Delta\Pi_{rigid})}{d\tau}\bigg|_{\tau=0} = \underbrace{\left(\frac{\partial \Pi}{\partial \bar{p}}\right)}_{>0} \cdot \underbrace{\left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau}\right)}_{>0 \text{ (by Lemma 1)}} > 0$$

This proves that the function is initially increasing.

2. We now analyze the limit of $\Delta\Pi_{rigid}(\tau)$ as $\tau \rightarrow \infty$. For any given producer price $p_j > 0$, the consumer price is $p_j^c = p_j(1 + \tau)$. As $\tau \rightarrow \infty$, $p_j^c \rightarrow \infty$. It is standard to assume $\lim_{p^c \rightarrow \infty} D(p_j^c, \bar{p}^c) = 0$, which in turn implies the quantity sold for any firm j converges to zero: $\lim_{\tau \rightarrow \infty} q_j = 0$.

The rigid firm charges the fixed price p_0 . Its profit is $\Pi_{rigid} = p_0 \cdot q_{rigid} - C(q_{rigid})$. As $\tau \rightarrow \infty$, $q_{rigid} \rightarrow 0$, and:

$$\lim_{\tau \rightarrow \infty} \Pi_{rigid}(\tau) = (p_0 \cdot 0) - C(0) = 0$$

The standard firm's profit Π^* must be non-negative (it can always choose $p_j = 0$, sell $q = 0$, and earn $\Pi = 0$). Since its profit is also choked off by the tax, its optimal profit must also converge to 0:

$$\lim_{\tau \rightarrow \infty} \Pi^*(\tau) = 0$$

Therefore, the limit of the profit difference is:

$$\lim_{\tau \rightarrow \infty} \Delta \Pi_{\text{rigid}}(\tau) = \lim_{\tau \rightarrow \infty} \Pi_{\text{rigid}}(\tau) - \lim_{\tau \rightarrow \infty} \Pi^*(\tau) = 0 - 0 = 0$$

In conclusion, we have a continuous function $\Delta \Pi_{\text{rigid}}(\tau)$ such that $\Delta \Pi_{\text{rigid}}(0) = 0$, $\frac{d}{d\tau} \Delta \Pi_{\text{rigid}}(0) > 0$, and $\lim_{\tau \rightarrow \infty} \Delta \Pi_{\text{rigid}}(\tau) = 0$. A function that starts at 0, increases, and must eventually return to 0 must be non-monotonic. $\square$

Proposition 8 (The Value of Flexibility). The profit advantage of a flexible firm over a rigid firm within the friction model, $\Delta \Pi_{\text{flex,rigid}} = \Pi_{\text{flex}} - \Pi_{\text{rigid}}$, is a strictly increasing function of the tax rate τ .

Proof. Let $V(\bar{p}_\infty, \tau) = \Pi_{\text{max}}(\bar{p}_\infty, \tau) - \Pi_{\text{rigid}}(\bar{p}_\infty, \tau)$ be the value of flexibility, where Π_{max} is the flexible firm's value function and $\Pi_{\text{rigid}} = \Pi(p_0, \bar{p}_\infty, \tau)$. We take the total derivative of V with respect to τ :

$$\frac{dV}{d\tau} = \left(\frac{\partial V}{\partial \bar{p}_\infty} \cdot \frac{d\bar{p}_\infty}{d\tau} \right) + \left(\frac{\partial V}{\partial \tau} \right)$$

We prove that both additive terms are positive.

1. **The Direct Effect** ($\frac{\partial V}{\partial \tau}$): By the Envelope Theorem, $\frac{\partial \Pi_{\text{max}}}{\partial \tau} = \frac{\partial \Pi}{\partial \tau} \Big|_{p_\infty^*}$, while $\frac{\partial \Pi_{\text{rigid}}}{\partial \tau} = \frac{\partial \Pi}{\partial \tau} \Big|_{p_0}$. Thus:

$$\frac{\partial V}{\partial \tau} = \frac{\partial \Pi}{\partial \tau} \Big|_{p_\infty^*} - \frac{\partial \Pi}{\partial \tau} \Big|_{p_0}$$

By Assumption 4 (Tax Effect), $\frac{\partial^2 \Pi}{\partial p_j \partial \tau} < 0$, so $\frac{\partial \Pi}{\partial \tau}$ is a decreasing function of p_j . Since $p_\infty^* < p_0$ (from Prop 1), it must be that $\frac{\partial \Pi}{\partial \tau} \Big|_{p_\infty^*} > \frac{\partial \Pi}{\partial \tau} \Big|_{p_0}$. This term is **positive**.

2. **The Environmental Effect** ($\frac{\partial V}{\partial \bar{p}_\infty} \cdot \frac{d\bar{p}_\infty}{d\tau}$): This term is a product of two components:

- $\frac{d\bar{p}_\infty}{d\tau}$: By Lemma 1 and Assumption 4, the producer price falls with the tax. This derivative is **negative**.
- $\frac{\partial V}{\partial \bar{p}_\infty} = \frac{\partial \Pi_{\text{max}}}{\partial \bar{p}_\infty} - \frac{\partial \Pi_{\text{rigid}}}{\partial \bar{p}_\infty} = \frac{\partial \Pi}{\partial \bar{p}} \Big|_{p_\infty^*} - \frac{\partial \Pi}{\partial \bar{p}} \Big|_{p_0}$. By Core Assumption 2 (Strategic Complementarity),

$\frac{\partial^2 \Pi}{\partial p_j \partial \bar{p}} > 0$, so $\frac{\partial \Pi}{\partial \bar{p}}$ is an increasing function of p_j . Since $p_\infty^* < p_0$, it must be that $\frac{\partial \Pi}{\partial \bar{p}}|_{p_\infty^*} < \frac{\partial \Pi}{\partial \bar{p}}|_{p_0}$. This derivative is **negative**.

The product of two negative terms is **positive**.

Since $\frac{dV}{d\tau}$ is the sum of two strictly positive terms, it is strictly positive. $\square$

Proposition 9 (Rigidity Bound). Define the profit difference of a rigid firm relative to a standard firm as

$$\Delta \Pi(\alpha_1) = \Pi_{rigid}(\alpha_1) - \Pi^*,$$

where α_1 denotes the share of rigid firms in the market. Suppose $\Delta \Pi(\alpha_1)$ is strictly increasing in α_1 with $\Delta \Pi(0) < 0$.

Then a rigid firm is better off than a standard firm if and only if $\alpha_1 > \alpha_1^*$, where the unique threshold $\alpha_1^* \in (0, 1)$ exists precisely when $\Delta \Pi(1) > 0$.

Proof. By continuity of the underlying demand and pricing functions, $\Delta \Pi(\alpha_1)$ is continuous. Moreover, only the market average price responds to α_1 , so

$$\frac{d\Delta \Pi}{d\alpha_1} = \frac{d\Pi_{rigid}}{d\alpha_1} = \frac{\partial \Pi}{\partial \bar{p}_\infty} \cdot \frac{d\bar{p}_\infty}{d\alpha_1}.$$

Assumption 3 implies $\frac{\partial \Pi}{\partial \bar{p}_\infty} > 0$, and Proposition 1 implies $\frac{d\bar{p}_\infty}{d\alpha_1} > 0$. Hence $\frac{d\Delta \Pi}{d\alpha_1} > 0$, so $\Delta \Pi(\alpha_1)$ is strictly increasing.

At $\alpha_1 = 0$, $\bar{p}_\infty = p^*$, and p^* is the unique maximizer of $\Pi(\cdot, p^*, \tau)$. Since $p_0 \neq p^*$,

$$\Delta \Pi(0) = \Pi(p_0, p^*, \tau) - \Pi(p^*, p^*, \tau) < 0.$$

We have a continuous, strictly increasing function with $\Delta \Pi(0) < 0$. By the Intermediate Value Theorem, a unique $\alpha_1^* \in (0, 1)$ such that $\Delta \Pi(\alpha_1^*) = 0$ exists if and only if $\Delta \Pi(1) > 0$.

If $\Pi_{rigid}(1) > \Pi^*$, such a root exists and strict monotonicity implies that

$$\Delta \Pi(\alpha_1) > 0 \iff \alpha_1 > \alpha_1^*.$$

$\square$

Proposition 10 (Effect of Tax on CTB). The difference in the consumer tax burden between the friction model and the standard model, $\Delta CTB = CTB_{friction} - CTB_{standard}$, is a strictly increasing function of the tax rate τ .

Proof. Define the consumer burden gap as a function of the tax rate. $\Delta CTB(\tau) = \bar{p}_{c,\infty} - p_c^* = \bar{p}_\infty(1 + \tau) - p^*(1 + \tau)$. Let $G_p(\tau) = \bar{p}_\infty(\tau) - p^*(\tau)$. Then $\Delta CTB(\tau) = (1 + \tau)G_p(\tau)$.

Using the product rule:

$$\frac{d(\Delta CTB)}{d\tau} = \underbrace{G_p(\tau)}_{> 0 \text{ from Prop. 1 and 3}} + (1 + \tau) \underbrace{\frac{dG_p}{d\tau}}_{\text{by Lemma 1}} > 0$$

□

Proposition 11 (The Effect of Rigidity on the CTB). The difference in the consumer tax burden, $\Delta CTB = CTB_{friction} - CTB_{standard}$, is a strictly increasing function of the proportion of rigid firms, α_1 .

Proof. Define the consumer burden gap as a function of the rigidity. $\Delta CTB(\alpha_1) = \bar{p}_{c,\infty}(\alpha_1) - p_c^*$. Differentiating with respect to α_1 (p_c^* is a constant w.r.t. α_1):

$$\frac{d(\Delta CTB)}{d\alpha_1} = \frac{d\bar{p}_{c,\infty}}{d\alpha_1} = \frac{d}{d\alpha_1}(\bar{p}_\infty(1 + \tau)) = (1 + \tau) \frac{d\bar{p}_\infty}{d\alpha_1}$$

which is positive as proved in Proposition 1

□

5.3.5 Dynamics

Proposition 12 (Dynamic Tax Incidence). Following the imposition of the tax at $t = 0$, and assuming a stable, monotonic price convergence (i.e., $\bar{p}_t < \bar{p}_{t-1}$ for all $t \geq 1$), the Consumer Tax Burden (CTB) is a monotonically decreasing function of time, while the Producer Tax Burden (PTB) is a monotonically increasing function of time.

Proof. The burdens at any time t are defined relative to the initial price p_0 and the average price at time t , $\bar{p}_t$ can be defined as follows.

$$PTB(0) = p_0 - p_0 \tag{23}$$

$$CTB(0) = p_0(1 + \tau) - p_0 = \tau p_0 \tag{24}$$

At the moment the tax is imposed, no prices have changed, so $\bar{p}_0 = p_0$.

$$PTB(t) = p_0 - \bar{p}_t$$

$$CTB(t) = \bar{p}_t(1 + \tau) - p_0$$

The Law of Motion governs the adjustment of $\bar{p}_t$ from its initial state p_0 to its new, lower long-run state $\bar{p}_\infty$ (where $\bar{p}_\infty < p_0$ as established in Proposition 1). Additionally, we assume the price path does not oscillate and is a strictly decreasing sequence:

$$\bar{p}_t < \bar{p}_{t-1} \quad \text{for } t \geq 1$$

Next, we compare tax burdens between periods t and $t - 1$. Comparison of $PTB(t)$ to $PTB(t - 1)$ yields,

$$PTB(t) = p_0 - \bar{p}_t$$

Since $\bar{p}_t < \bar{p}_{t-1}$, it follows that $-\bar{p}_t > -\bar{p}_{t-1}$.

$$p_0 - \bar{p}_t > p_0 - \bar{p}_{t-1} \implies PTB(t) > PTB(t - 1)$$

The Producer Tax Burden is monotonically increasing from its minimum of 0.

Comparison of $CTB(t)$ to $CTB(t - 1)$ yields,

$$CTB(t) = \bar{p}_t(1 + \tau) - p_0$$

Since $\bar{p}_t < \bar{p}_{t-1}$, it follows that $\bar{p}_t(1 + \tau) < \bar{p}_{t-1}(1 + \tau)$.

$$\bar{p}_t(1 + \tau) - p_0 < \bar{p}_{t-1}(1 + \tau) - p_0 \implies CTB(t) < CTB(t - 1)$$

The Consumer Tax Burden is monotonically decreasing from its maximum of τp_0 . Therefore, we can conclude that consumers bear the maximum burden at $t = 0$, and this burden is gradually shifted to producers as time progresses and firms adjust their prices.

□

Proposition 13 (The Source of Dynamic Persistence). The model's dynamic "memory" (i.e., its dependence on prices before $t - 1$) is generated exclusively by the presence of partially flexible (Type 2) firms. In the absence of Type 2 firms ($\alpha_2 = 0$), the model's dynamics simplify to a first-order process.

Proof. The average price $\bar{p}_t$ is the weighted average of all firm types:

$$\bar{p}_t = \alpha_1 p_0 + \alpha_2 \bar{p}_{2,t} + \alpha_3 p_t^*$$

where $p_t^* = p^*(\bar{p}_{t-1}, \tau)$ is the optimal price. $\bar{p}_{2,t}$ is the average price of Type 2 (Calvo) firms. By definition, a fraction μ_2 of them update to p_t^* , while the remaining $(1 - \mu_2)$ fraction keep their price

from $t - 1$.

$$\bar{p}_{2,t} = (1 - \mu_2)\bar{p}_{2,t-1} + \mu_2 p_t^*$$

This recursive definition means that $\bar{p}_{2,t}$ is a weighted average of *all past optimal prices* ($p_t^*, p_{t-1}^*, p_{t-2}^*, \dots$). This is what creates the higher-order dynamics (persistence). If there are no Calvo firms, the Law of Motion becomes:

$$\bar{p}_t = \alpha_1 p_0 + \alpha_3 p_t^* \quad (\text{where } \alpha_1 + \alpha_3 = 1)$$

Substituting the best-response function:

$$\bar{p}_t = \alpha_1 p_0 + \alpha_3 p^*(\bar{p}_{t-1}, \tau)$$

This is a first-order difference equation. $\bar{p}_t$ depends *only* on $\bar{p}_{t-1}$. All "memory" of $\bar{p}_{t-2}$ and earlier is lost.

□

Proposition 14 (Rigidity and Convergence Speed). A higher proportion of rigid firms (α_1) slows the market's convergence to its long-run equilibrium.

Proof. The market's dynamic adjustment is driven by the mass of non-rigid firms (Types 2 and 3), which has a total measure of $(1 - \alpha_1)$. These are the only firms that can update their prices in response to the tax and move the average price $\bar{p}_t$ toward its new steady state $\bar{p}_\infty$.

A higher proportion of rigid firms α_1 mechanically reduces the mass of adjusting firms $(1 - \alpha_1)$.

This makes the aggregate price $\bar{p}_t$ less responsive to any deviation from the steady state. In the Law of Motion (Equation 7), the coefficients on the dynamic terms (e.g., $p^*(\bar{p}_{t-1}, \tau)$) are all scaled by $(\alpha_2 + \alpha_3) = (1 - \alpha_1)$. A larger α_1 reduces the magnitude of these terms, dampening the system's response.

A less responsive system, by definition, converges more slowly to its long-run equilibrium following a shock.

□

Proposition 15 (Consumer Price Overshooting). Following the tax, the average consumer price $\bar{p}_t^c$ overshoots its new long-run equilibrium. It jumps to its highest point at $t = 0$ and then (assuming monotonic convergence) decreases to its final steady state, $\bar{p}_{c,\infty}$.

Proof. The proof consists of two parts. We first compare the start and end points of the price path, and then characterize the path between them.

Step 1: The consumer price at a fixed time t is $\bar{p}_t^c = \bar{p}_t(1 + \tau)$.

At $t = 0$, no prices have adjusted, so $\bar{p}_0 = p_0$. The consumer price instantly becomes:

$$\bar{p}_0^c = p_0(1 + \tau)$$

In the long-run steady state ($t \rightarrow \infty$), the producer price is $\bar{p}_\infty$. The consumer price is:

$$\bar{p}_{c,\infty} = \bar{p}_\infty(1 + \tau)$$

From the long-run aggregation equation (Eq. 13), $\bar{p}_\infty$ is a weighted average of p_0 and p_∞^* . From the proof of Proposition 1, we know $p_\infty^* < p_0$. Therefore, it must be that $\bar{p}_\infty < p_0$.

Since $\bar{p}_\infty < p_0$, it follows that the final consumer price is strictly lower than the initial consumer price:

$$\bar{p}_{c,\infty} < \bar{p}_0^c$$

This proves the initial consumer price is strictly higher than the final long-run consumer price, meaning the price must, in net, fall to reach its new equilibrium.

Step 2: To prove the path is monotonically decreasing, we must characterize the sequence $\{\bar{p}_t^c\}_{t=1}^\infty$. We make the standard assumption that the parameters of the model (A.1-A.4) are such that the dynamic system converges to its steady state monotonically, without overshooting or oscillating. This means the producer price path is a strictly decreasing sequence:

$$\bar{p}_t < \bar{p}_{t-1} \quad \text{for all } t \geq 1$$

Given this, it follows directly that the consumer price path is also monotonically decreasing for all $t \geq 1$:

$$\bar{p}_t(1 + \tau) < \bar{p}_{t-1}(1 + \tau) \implies \bar{p}_t^c < \bar{p}_{t-1}^c$$

Thus, the average consumer price jumps to its highest point at $t = 0$ and then monotonically decreases to its final steady state. $\square$

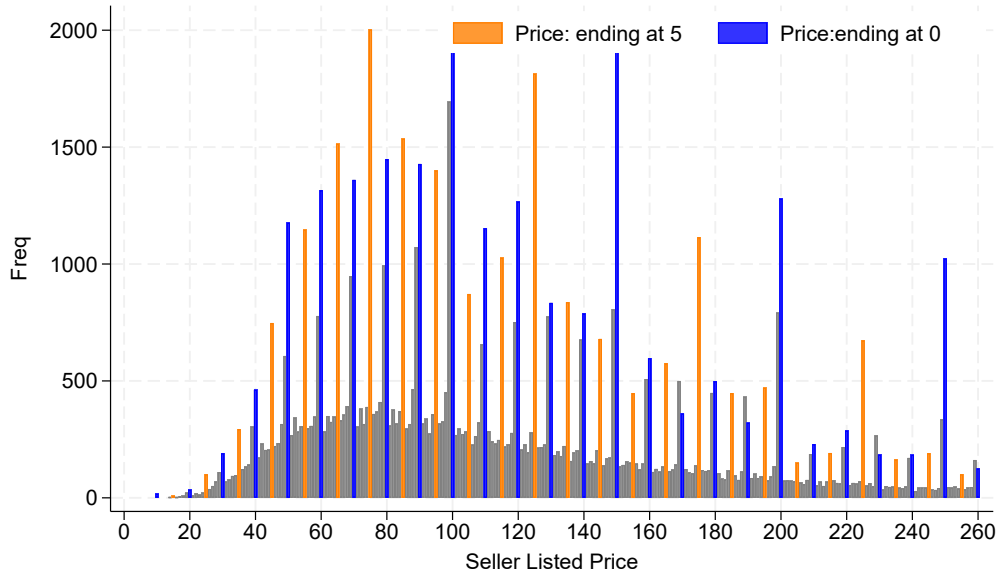
6 Conclusion

While it is widely accepted that consumer optimization frictions affect tax incidence, far less is known about whether and how supplier frictions change relative burdens. We attempt to mitigate this gap by studying producer frictions in a market where they are likely to be present: short-term listings on a large platform firm. We find that Airbnb host pricing behaviors exhibit significant

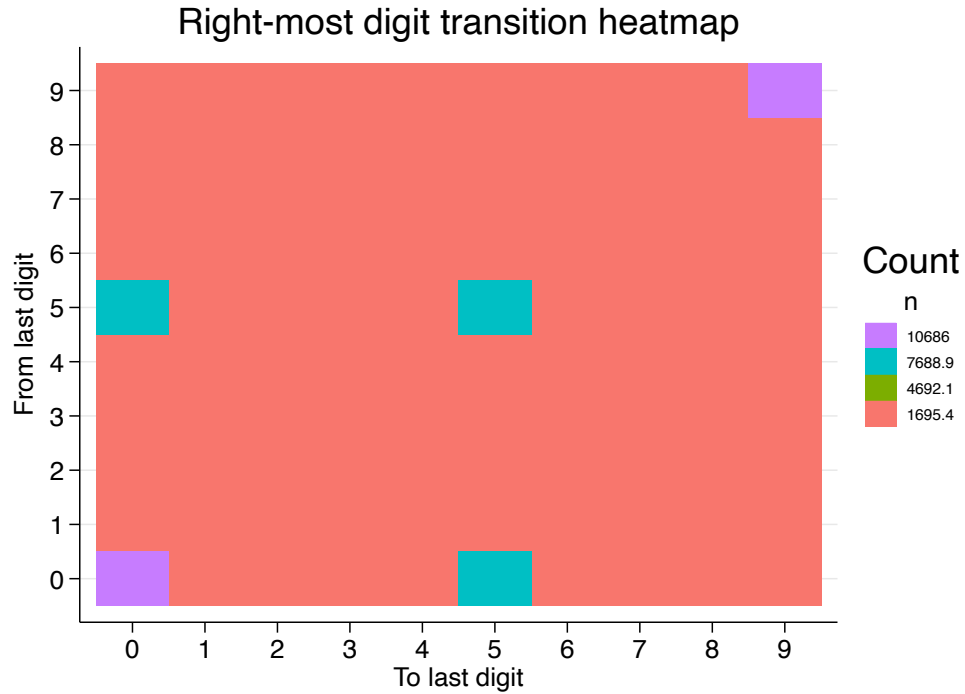
frictions and that these frictions are not uniformly distributed but correlate with other measures of professionalism.

Using a natural experiment that varied the effective tax rate, we find that those hosts with a low correlation with local hotel prices before the policy passed on more of the tax burden to consumers. We build a simple model to understand the welfare consequences of our empirical findings. While we find that deadweight loss is exacerbated by the presence of supplier frictions, producer surplus for hosts facing all levels of friction is higher compared with a frictionless world.

FIGURE 1: Airbnb Host Pricing Heuristics



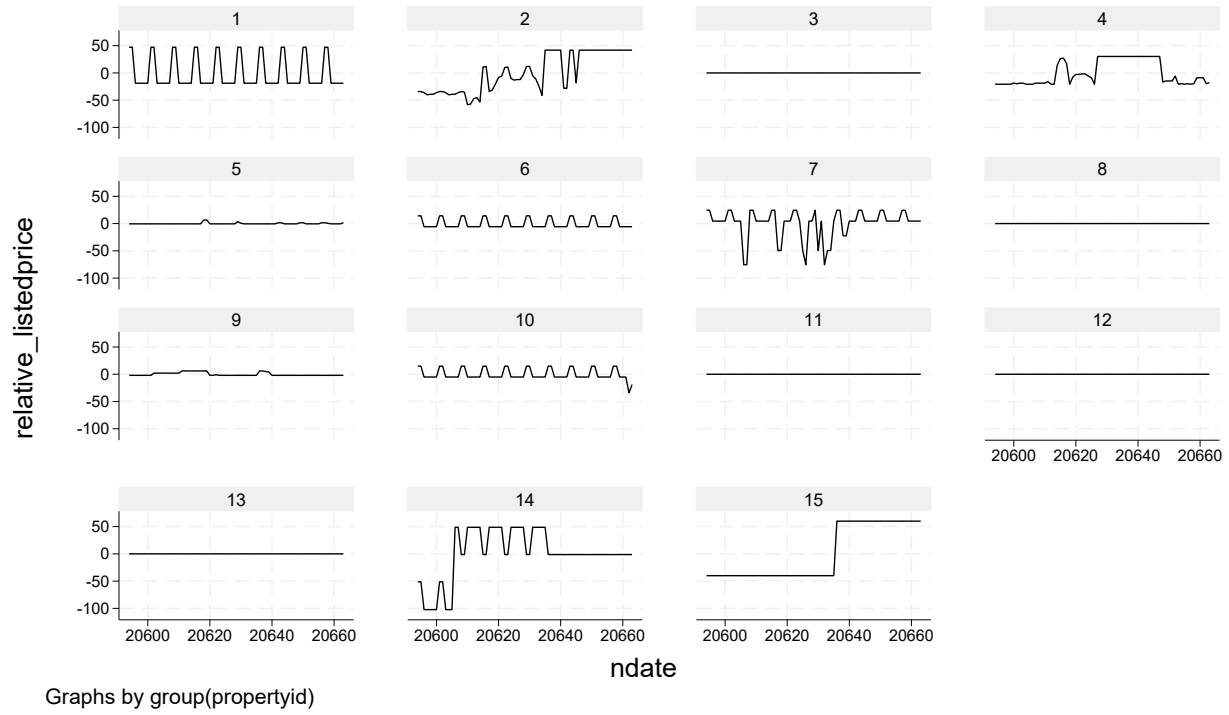
(a) Listing Prices Prior to VCA Adoption



(b) Right-most Digit Transition Matrix

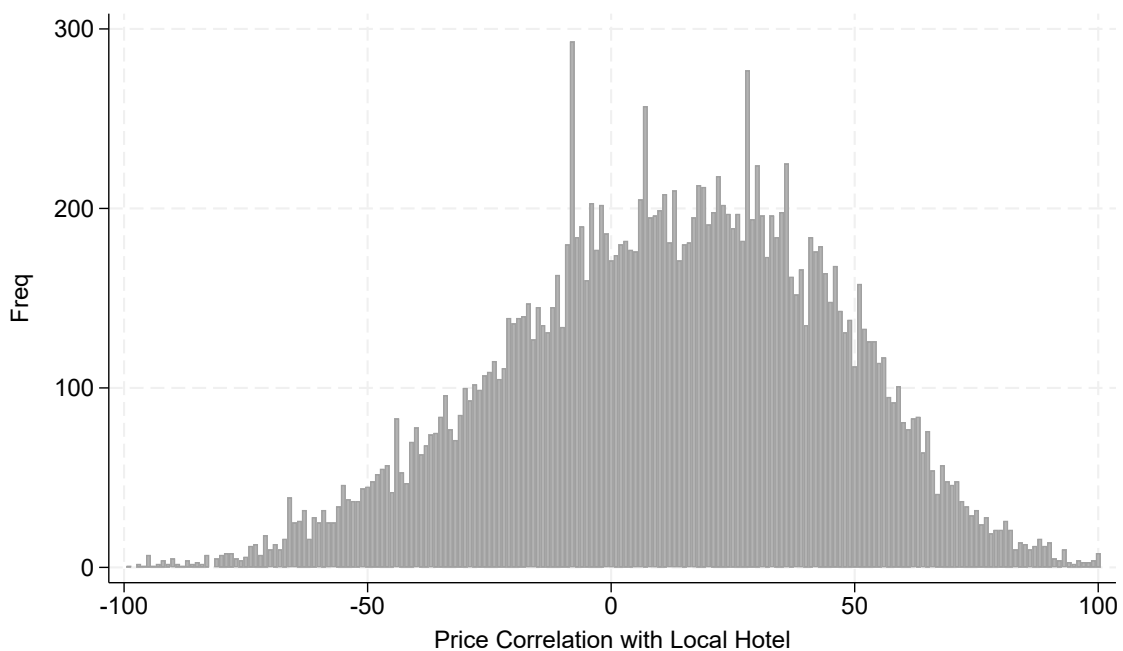
Notes: For both panels, data include all listed prices prior to VCA adoption. In panel (a) Blue bars represent prices divisible by ten; Orange bars represent prices divisible by five. In panel (b), each square in the matrix represents a count of price transitions between prices endings in specific digits. Specifically, the count is of price changes where the initial price ended in the vertical axis and changed to a price that ended on the digit on the horizontal axis, i.e. a price change from \$100 to \$120 would be counted in the bottom most and left most square as both prices end with zero.

FIGURE 2: Airbnb Host Pricing Over Time



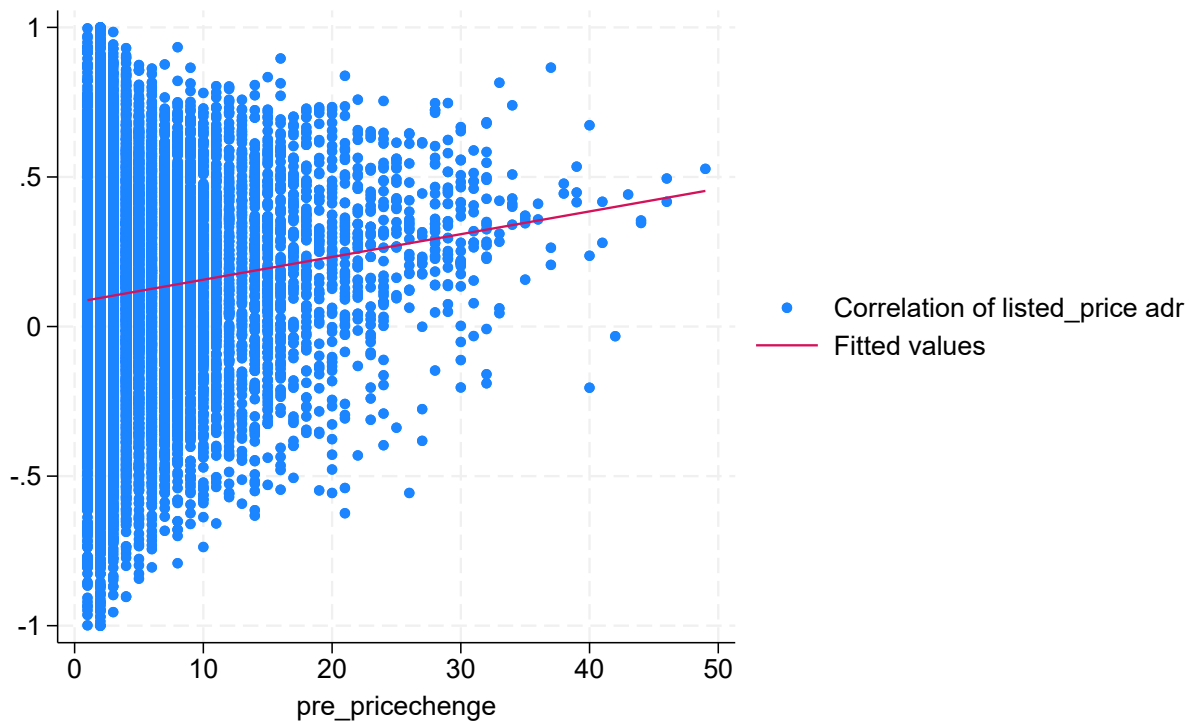
Notes: Each panel displays the pre-policy pricing trajectory of sixteen randomly selected properties from our analysis dataset. Figures 3, 8, 11, 12, and 13 exhibit a constant price over time, implying that their hosts do not employ any kind of dynamic pricing strategy— profit maximizing only if demand was static. Figures 1, 6 and 14 exhibit a simplified dynamic strategy in which price fluctuates between two values; one price for weekdays and the second price for weekend days.

FIGURE 3: Distribution of Correlations between Airbnb Properties and Local Hotel Prices



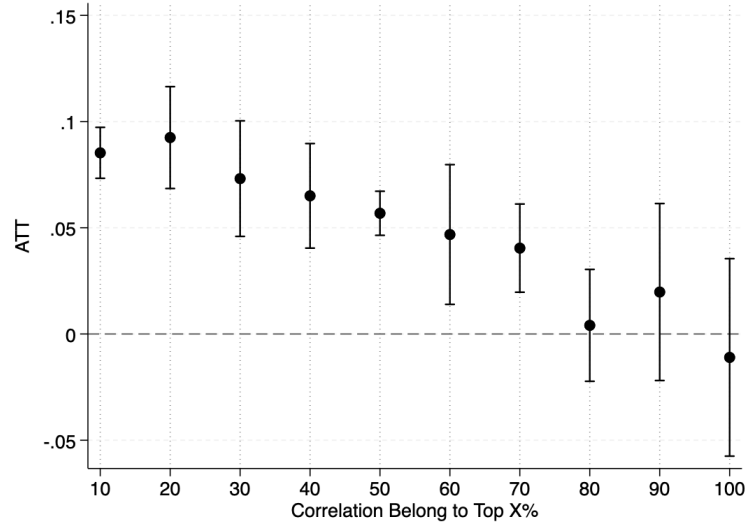
Notes: This histogram plots the distribution of pre-policy price correlations between Airbnb properties and local hotel prices for properties located in treated cities. Positive correlation values indicate the Airbnb property listing price moved with local hotel pricing (i.e. were responsive to local demand conditions).

FIGURE 4: Relationship between Hotel Price Correlations and Price Change After Policy



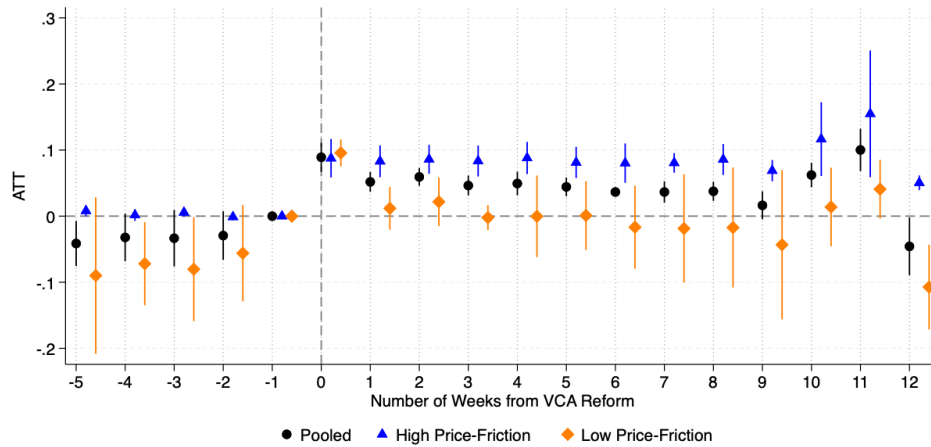
Notes: This scatter plot shows the relationship between pre-policy hotel price correlations and the absolute percentage price change before and after the policy. The line of best fit slopes upward, suggesting properties that moved together with hotel prices prior to the policy were more likely to respond to the VCA by changing their price.

FIGURE 5: The Effects of VCA on log consumer paid price: Property with Corr Percentile X%



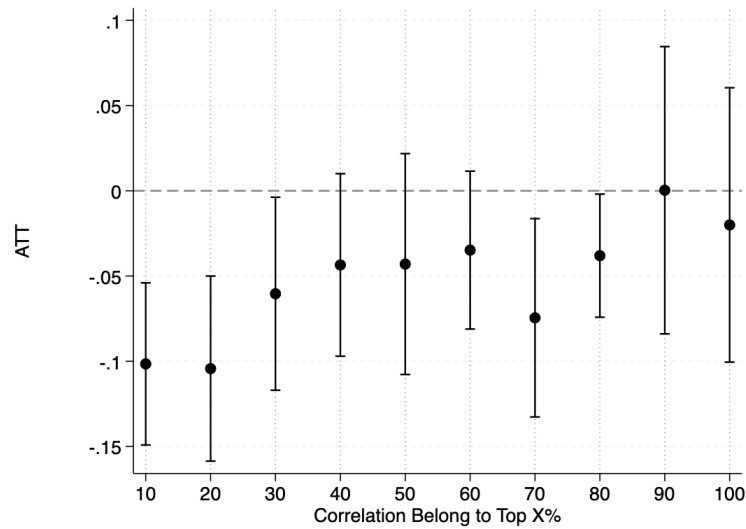
The Figures plot the average treatment effect on the treated (ATT) regression coefficient estimates using [Callaway and Sant'Anna \(2021\)](#) estimator from Column (5). The dependent variable is the log of consumer paid price, defined as the ultimate price paid by the consumer, which includes the booked price plus the cleaning fee and the service fee. Note that the observed booking price should be tax inclusive but tax exclusive. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. All the regression are clustered at MSA level. The bars indicate 95% confidence intervals with standard errors clustered at the MSA level.

FIGURE 6: Event Study: log consumer booked price



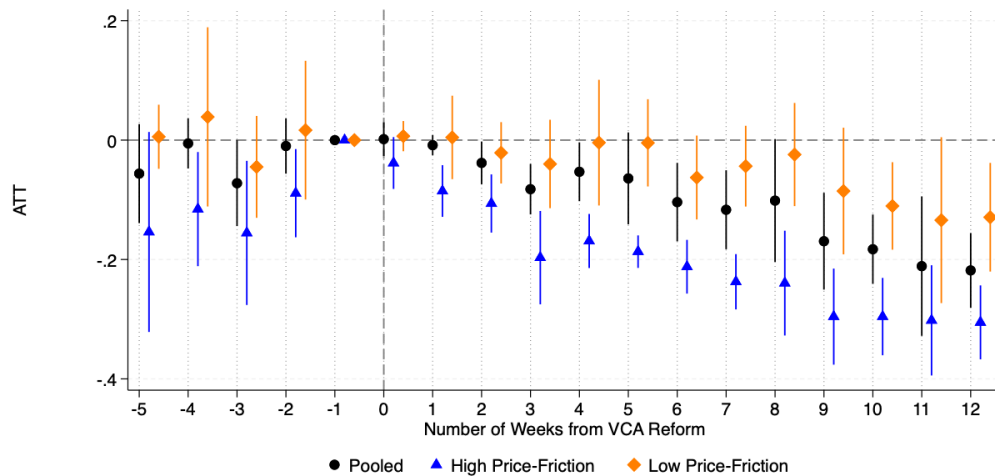
This table reports the aggregated average treatment effect on the treated (ATT) regression coefficient estimates using [Callaway and Sant'Anna \(2021\)](#) estimator. Following [Roth \(2024\)](#), we use the long-difference for the pre- and post-treatment coefficients to better interpret the pre-trends relative to the reference period ($t=-1$). The dependent variable is the log of consumer paid price, defined as the ultimate price paid by the consumer, which includes the booked price plus the cleaning fee and the service fee. Note that the observed booking price should be tax inclusive but tax exclusive. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. All the regressions are clustered at the MSA level. The bars indicate 95% confidence intervals with standard errors clustered at the MSA level.

FIGURE 7: The Effects of VCA on Log Reservation Ratio: Property with Corr X%



The Figures plot the average treatment effect on the treated (ATT) regression coefficient estimates using [Callaway and Sant'Anna \(2021\)](#) estimator. Following [Roth \(2024\)](#), we use the long-difference for the pre- and post-treatment coefficients to better interpret the pre-trends relative to the reference period ($t=-1$). The dependent variable is the log of consumer paid price, defined as the ultimate price paid by the consumer, which includes the booked price plus the cleaning fee and the service fee. Note that the observed booking price should be tax inclusive but tax exclusive. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. All the regression are clustered at MSA level. The bars indicate 95% confidence intervals with standard errors clustered at the MSA level.

FIGURE 8: Event Study: log Reservation Ratio



This table reports the aggregated average treatment effect on the treated (ATT) regression coefficient estimates using [Callaway and Sant'Anna \(2021\)](#) estimator. Following [Roth \(2024\)](#), we use the long-difference for the pre- and post-treatment coefficients to better interpret the pre-trends relative to the reference period ($t=-1$). The dependent variable is the log of consumer paid price, defined as the ultimate price paid by the consumer, which includes the booked price plus the cleaning fee and the service fee. Note that the observed booking price should be tax inclusive but tax exclusive. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. All the regressions are clustered at the MSA level. The bars indicate 95% confidence intervals with standard errors clustered at the MSA level.

TABLE 1: Descriptive Statistics

	All Property			High Price Friction Group			Low Price Friction Group		
	mean	sd	p50	mean	sd	p50	mean	sd	p50
Panel A: Treated Cities									
% Multiple Host	0.46	0.49	0.00	0.51	0.49	0.80	0.40	0.48	0.00
% Entire House	0.63	0.48	1.00	0.67	0.47	1.00	0.59	0.49	1.00
No. bathrooms	1.22	0.47	1.00	1.22	0.46	1.00	1.21	0.48	1.00
No. bedrooms	1.14	0.72	1.00	1.16	0.73	1.00	1.13	0.70	1.00
Listed price	126.21	70.18	106.83	131.04	70.25	113.17	120.11	69.63	100.00
Consumer Paid Price	172.62	97.04	150.00	181.33	98.23	161.32	161.62	94.38	136.72
strict_R	4.18	2.19	4.80	4.40	2.09	5.00	3.91	2.29	4.20
r_ratio	0.63	0.32	0.74	0.66	0.30	0.77	0.59	0.33	0.66
Average Tax Rate Change	0.12	0.03	0.14	0.12	0.03	0.14	0.12	0.03	0.14
#Property		5658			3159			2499	
Obs									
Panel B: Control Cities									
	mean	sd	p50	mean	sd	p50	mean	sd	p50
% Multiple Host	0.22	0.41	0.00	0.16	0.36	0.00	0.26	0.43	0.00
% Entire House	0.68	0.47	1.00	0.62	0.48	1.00	0.72	0.45	1.00
No. bathrooms	1.33	0.58	1.00	1.33	0.58	1.00	1.33	0.58	1.00
No. bedrooms	1.49	0.89	1.00	1.46	0.87	1.00	1.51	0.90	1.00
Listed price	166.41	165.23	116.43	183.25	165.67	125.00	155.23	164.03	113.97
Consumer Paid Price	189.63	150.91	145.00	176.88	156.75	125.00	193.68	148.82	150.00
strict_R	1.68	2.14	0.60	0.90	1.80	0.00	2.20	2.19	1.60
r_ratio	0.25	0.32	0.09	0.13	0.26	0.00	0.33	0.33	0.24
#Property		3465			1382			2083	

Notes: This table presents descriptive statistics for treated cities (Boulder, Philadelphia, Phoenix, San Diego, and Los Angeles) and control cities (Austin, Dallas, Houston). We further break down these statistics by high and low optimization friction properties, where "High Friction" refers to properties whose pre-policy price correlation was below the median and "Low Friction" refers to those that are above.

TABLE 2: Staggered DID Estimation: Log Consumer Paid Price

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
VCA Reform	.0816*** (.0088)	.0908*** (.0066)	.0777*** (.0036)	.0878*** (.0091)	.0700*** (.0175)	.0464*** (.0054)
	292222	292222	271772	271772	273732	143779
Baseline Mean						
Average Tax Rate Change			.121			
Panel B: High Price Friction Group: Corr $\leq$ Median						
VCA Reform	.0952*** (.0128)	.1002*** (.0106)	.1143*** (.0013)	.1143*** (.0013)	.1104*** (.0145)	.0764*** (.0122)
	131672	131672	121084	121084	121138	60441
Baseline Mean						
Average Tax Rate Change			.121			
Panel C: Low Price Friction Group: Corr $>$ Median						
VCA Reform	.0698*** (.0083)	.0824*** (.0063)	.0483*** (.0045)	.0662*** (.0171)	.0370 (.0246)	.0196*** (.0074)
	160550	160550	150688	150688	152594	83338
Baseline Mean						
Average Tax Rate Change			.121			
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	✓	✓	✓	✓	✓	✓
Property FE	✓	✓	✓	✓	✓	✓
Cities-Weekly Demand		✓		✓		✓
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and Columns (5) and (6) use an estimator following [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

TABLE 3: Staggered DID Estimation: Log Reservation Ratio

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
VCA Reform	-.066*** (.0214) 292222	-.043** (.0200) 292222	-.113*** (.0075) 271772	-.077** (.0316) 271772	-.105*** (.0326) 273732	-.050* (.0272) 143779
Panel B: High Price Friction Group: Corr $\leq$ Median						
VCA Reform	-.072*** (.0188) 131672	-.047** (.0189) 131672	-.123*** (.0069) 121084	-.092*** (.0249) 121084	-.124*** (.0273) 121138	-.069** (.0282) 60441
Panel C: Low Price Friction Group: Corr $>$ Median						
VCA Reform	-.063*** (.0234) 160550	-.039* (.0211) 160550	-.102*** (.0087) 150688	-.063 (.0385) 150688	-.091** (.0358) 152594	-.034 (.0262) 83338
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	✓	✓	✓	✓	✓	✓
Property FE	✓	✓	✓	✓	✓	✓
Cities-Weekly Demand		✓		✓		✓
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and Columns (5) and (6) use an estimator following [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

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Online Appendix: For Online Publication

Section A	Additional Tables and Figures
Section B	Balanced Panel Estimation
Section C	Daily Unit Estimation
Section D	Robustness of Price Friction

A Additional Tables and Figures

TABLE A1: Policy Table

	#properties	#observations	average pre-policy listed price	Tax Rate	Policy Date
Boulder	868	3847	132.90	7.5	10/1/2016
Philadelphia	1,208	5345	110.40	8.5	7/15/2015
Phoenix	244	1124	99.46	5	7/1/2015
SanDiego	2,229	9548	149.38	10.5	7/15/2015
LosAngeles	8,419	36820	123.40	14	8/1/2016
Austin	4,479	19754	185.93		
Dallas	799	3508	104.74		
Houston	1,402	6179	95.28		

TABLE A2: Staggered DID Estimation: Log Listed Price

	(1)	(2)	(3)	(4)
Panel A: Pooled				
VCA Reform	-.035*** (.0094)	-.020** (.0091)	-.036* (.0193)	-.054*** (.0067)
Obs	164214	164214	114912	143779
Baseline Mean				
Average Tax Rate Change	.121	.121	.1212	.121
Panel B: High Price Friction Group: Corr ≤ Median				
VCA Reform	-.017** (.0086)	-.009 (.0076)	-.009*** (.0036)	-.014*** (.0055)
Obs	69858	69858	48716	60441
Baseline Mean				
Average Tax Rate Change	.121	.121	.1212	.121
Panel C: Low Price Friction Group: Corr > Median				
VCA Reform	-.050*** (.0130)	-.030** (.0121)	-.062* (.0323)	-.089*** (.0201)
Obs	94356	94356	66196	83338
Baseline Mean				
Average Tax Rate Change	.121	.121	.1212	.121
Estimator	TWFE	TWFE	Multiplgt	CSDID
Year-Week FE	√	√	√	√
Property FE	√	√	√	√
Cities-Weekly Demand		√	√	√
Cluster	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log listed price set by the host. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

TABLE A3: Staggered DID Estimation: Log Consumer Paid Price

	(1)	(2)	(3)	(4)
Panel A: Pooled				
VCA Reform	.0654*** (.0093)	.0766*** (.0071)	.0672*** (.0082)	.0477*** (.0096)
Obs	94800	94800	59162	62060
Baseline Mean				
Average Tax Rate Change	.121	.121	.1212	.121
Panel B: High Price Friction Group: Corr $\leq$ Median				
VCA Reform	.0790*** (.0101)	.0870*** (.0084)	.0849*** (.0016)	.0794*** (.0068)
Obs	33920	33920	20656	28826
Baseline Mean				
Average Tax Rate Change	.121	.121	.1212	.121
Panel C: Low Price Friction Group: Corr $>$ Median				
VCA Reform	.0557*** (.0099)	.0685*** (.0078)	.0559*** (.0127)	.0204*** (.0070)
Obs	60880	60880	38506	56073
Baseline Mean				
Average Tax Rate Change	.121	.121	.1212	.121
Estimator	TWFE	TWFE	Multiplegt	CSDID
Year-Week FE	✓	✓	✓	✓
Property FE	✓	✓	✓	✓
Cities-Weekly Demand		✓	✓	✓
Cluster	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

B Balanced Panel Estimation

TABLE A4: Staggered DID Estimation: Log Consumer Paid Price

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
VCA Reform	.0777*** (.0095) 164214	.0894*** (.0070) 164214	.0749*** (.0006) 114912	.0887*** (.0139) 114912	.0690*** (.0167) 164214	.0452*** (.0101) 84969
Panel B: High Price Friction Group: Corr $\leq$ Median						
VCA Reform	.0902*** (.0132) 69858	.0965*** (.0105) 69858	.1108*** (.0007) 48716	.1084*** (.0032) 48716	.1081*** (.0132) 69858	.0668*** (.0148) 34785
Baseline Mean						
Average Tax Rate Change			.121			
Panel C: Low Price Friction Group: Corr $>$ Median						
VCA Reform	.0676*** (.0088) 94356	.0826*** (.0063) 94356	.0473*** (.0006) 66196	.0701*** (.0236) 66196	.0391** (.0198) 94356	.0275*** (.0052) 50184
Baseline Mean						
Average Tax Rate Change			.121			
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	✓	✓	✓	✓	✓	✓
Property FE	✓	✓	✓	✓	✓	✓
Cities-Weekly Demand		✓		✓		✓
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and Columns (5) and (6) use an estimator following [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

TABLE A5: Staggered DID Estimation: Log Reservation Ratio

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
VCA Reform	-.070*** (.0197) 164214	-.043** (.0193) 164214	-.091*** (.0014) 114912	-.045 (.0482) 114912	-.098*** (.0308) 164214	-.037 (.0297) 84969
Panel B: High Price Friction Group: Corr $\leq$ Median						
VCA Reform	-.066*** (.0235) 94356	-.040* (.0216) 94356	-.080*** (.0016) 66196	-.030 (.0525) 66196	-.087** (.0354) 94356	-.021 (.0340) 50184
Panel C: Low Price Friction Group: Corr $>$ Median						
VCA Reform	-.074*** (.0147) 69858	-.047*** (.0168) 69858	-.106*** (.0014) 48716	-.066* (.0403) 48716	-.112*** (.0238) 69858	-.057* (.0330) 34785
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	✓	✓	✓	✓	✓	✓
Property FE	✓	✓	✓	✓	✓	✓
Cities-Weekly Demand		✓		✓		✓
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and Columns (5) and (6) use an estimator following [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

C Robustness of Price Friction

TABLE A6: Staggered DID Estimation: Log Listed Consumer Booked Price

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Without Instant-book-Feature						
VCA Reform	.0842*** (.0085)	.0926*** (.0065)	.0663*** (.0070)	.0890*** (.0084)	.0729*** (.0166)	.0482*** (.0051)
	234656	234656	48815.	222957	218724	121368
Panel B: With Instant-book-Feature						
VCA Reform	.0690*** (.0099)	.0824*** (.0069)	.0800*** (.0033)	.0821*** (.0126)	.0547*** (.0191)	.0372*** (.0083)
	57566	57566	222957	48815	55008	22411
Panel C: Single Host						
VCA Reform	.0865*** (.0108)	.0940*** (.0084)	.0854*** (.0019)	.0911*** (.0052)	.0787*** (.0143)	.0422*** (.0047)
	147431	147431	139052	139052	137066	64513
Panel D: Multiple Host						
VCA Reform	.0763*** (.0085)	.0874*** (.0067)	.0705*** (.0060)	.0865*** (.0145)	.0615*** (.0198)	.0499*** (.0067)
	144791	144791	132720	132720	136666	79266
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	✓	✓	✓	✓	✓	✓
Property FE	✓	✓	✓	✓	✓	✓
Cities-Weekly Demand		✓		✓		✓
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and Columns (5) and (6) use an estimator following [Callaway and Sant'Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

TABLE A7: Staggered DID Estimation: Log Reservation Ratio

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Without Instant-book-Feature						
VCA Reform	-.072*** (.0222) 234656	-.048** (.0202) 234656	-.096*** (.0097) 48815	-.084*** (.0298) 222957	-.110*** (.0365) 218724	-.053** (.0270) 121368
Panel B: With Instant-book-Feature						
VCA Reform	-.048** (.0192) 57566	-.024 (.0203) 57566	-.118*** (.0081) 222957	-.052 (.0430) 48815	-.086*** (.0260) 55008	-.034 (.0241) 22411
Panel C: Single Host						
VCA Reform	-.067*** (.0234) 147431	-.042** (.0217) 147431	-.117*** (.0057) 139052	-.085*** (.0270) 139052	-.109*** (.0307) 137066	-.033 (.0270) 64513
Panel D: Multiple Host						
VCA Reform	-.067*** (.0193) 144791	-.043** (.0181) 144791	-.107*** (.0101) 132720	-.068* (.0361) 132720	-.100*** (.0314) 136666	-.062*** (.0235) 79266
Estimator	TWFE	TWFE	Multiplengt	Multiplengt	CSDID	CSDID
Year-Week FE	✓	✓	✓	✓	✓	✓
Property FE	✓	✓	✓	✓	✓	✓
Cities-Weekly Demand		✓		✓		✓
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following [Sun and Abraham \(2021\)](#) and Columns (5) and (6) use an estimator following [Callaway and Sant’Anna \(2021\)](#). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

D Daily Unit Estimation